

GOWTHAM KANDASAMY

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PROFESSIONAL SUMMARY

Senior Software Engineer with 7 years of engineering experience specializing in backend architecture, high-impact workflow automation, and technical leadership. Proven track record of reclaiming business capacities and achieving \$13k/year in infrastructure cost reductions through architectural innovation. Expert across Python and Java ecosystems, leveraging AI-assisted workflows (Cursor AI), Model Context Protocol (MCP), and Retrieval-Augmented Generation (RAG) to accelerate feature velocity and deploy containerized enterprise services.

CORE COMPETENCIES

AI&Architecture:	Retrieval-Augmented Generation (RAG), Vector Embeddings, pgvector, Google Gemini, Model Context Protocol (MCP), Cursor AI, Ollama, Docling
Languages & Frameworks:	Python (FastAPI, SQLAlchemy, Hamilton), Java 17 (Spring Boot, Hibernate), TypeScript, React 19, Vite, Tailwind CSS, Streamlit, PostgreSQL, MySQL, ChromaDB, Alembic
Automation & DevOps:	Docker, Docker Compose, Kubernetes (EKS), Helm, AWS (S3, Secrets Manager), CI/CD, RPA (UiPath), Selenium, PyMuPDF, PDFlib TET, WebSockets, Google Apps Script
Security & Integrations:	JWT, bcrypt, Okta, Azure AD, Dropbox API, Microsoft Teams Webhooks, OpenCSV, REST APIs

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Perch Energy (Helios Arcadia New Energy) Mar 2025 – May 2026 | Remote

- Architected a centralized automation hub that slashed manual processing time by 85%, streamlining critical ingestion layer for 18 global utilities across 198 concurrent projects.
- Implemented a Hamilton-based DAG methodology for dynamic data processing, seamlessly managing scalable subscriber profiles for over 60,000 accounts.
- Accelerated engineering velocity by 50% by designing context-aware cursor rule and local tool mappings that automated codebase indexing, enabling engineers to instantly generate rapid, high-precision utility components.

Senior Software Engineer – Arcadia (Urjanet, an Arcadia enterprise) May 2022 – Mar 2025 | Remote, India

- Engineered a high-performance Java 17 contract platform (CAE) featuring an advanced three-stage Auto-QC engine for structural PDF text-layer diffing, reclaiming 3 FTE capacities daily.
- Won the 2024 Corporate Hackathon by single-handedly migrating legacy toolsets to a free, open-source PyMuPDF solution, yielding \$13k/year in recurring operations cost savings.
- Built the flagship TDS engine handling 140+ global LSEs, utilizing automated text extraction frameworks with Selenium and PDFlib TET to maximize precision.
- Selected by management to pioneer technical screening initiatives; successfully interviewed, hired, and onboarded high-performance engineering squads to scale production.

Software Engineer – Urjanet Energy Solutions May 2019 – May 2022 | Chennai, India

- Standardized robust regional layout schemas for over 50 utilities utilizing structured Java and Hibernate backend parsing components.
- Automated engineering JIRA triage and complex metrics analysis by creating custom integration features using REST APIs within the internal anomaly detector platform (PAD).
- Created and published custom internal Google Apps Script functions to automate sheet tracking across operational cross-functional squads.

PERSONAL TECHNICAL PROJECTS

Claims Auditor AI – Enterprise Health Insurance Claims Auditing Platform 2026

FastAPI · React 19 · TypeScript · PostgreSQL · pgvector · Gemini 2.5 Flash · Docling · Docker Compose · JWT · WebSockets · Alembic

- Architected a full-stack AI auditing platform: policy PDF ingestion via Docling, heading-aware chunking, Gemini embeddings (3072-dim) stored in pgvector, and a React 19 dashboard with live WebSocket progress (Policy Ingestion, Claims Desk, Audit Console, RAG Chat views).
- Built a Gemini 2.5 Flash cognitive audit engine that compares claim vs. policy Markdown and returns structured APPROVED / FLAGGED / DENIED verdicts with policy citations and violation severity.
- Delivered JWT multi-tenant auth, user-scoped repositories, Alembic migrations, async 202-Accepted background pipelines, and real-time WebSocket status streams.

AI Personal Habit Tracker & Assistant Coach 2026

FastAPI · PostgreSQL · Streamlit · FastMCP (SSE) · ChromaDB · Ollama · Docker Compose

- Containerized habit tracker with an MCP Server over SSE for deterministic agent-driven database operations and a local RAG layer (ChromaDB, all-minilm:v2, 384-dim) injecting historical trends into the system prompt.
- Integrated repository hooks to keep relational data and vector memory in sync; persistent chatbot sidebar available across all Streamlit views.

EDUCATION & HONORS

B.E. Electronics & Communication Engineering – Karpagam College of Engineering 2015 – 2019 | Score: 86.63%

12th & 10th Grade (State Board) – Swamy Vivekanandha Matric Hr. Sec. School 2013 – 2015 | Score: 92.1% / 92.6%

Honors: Recipient of Best Project of the Year (CAE), "It's My Ship" Award, and multiple quarterly engineering excellence milestones.